Yuxuan Liu

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BUPT Undergraduate

Graph-Enhanced RAG

Knowledge Graph Quality

Multimodal Document Understanding

Education

Beijing University of Posts and Telecommunications (BUPT)

Sept. 2023 – Jun. 2027 (Expected)

B.Eng. in Telecommunications Engineering GPA: 82.05/100 | CET-6: 560

Relevant courses: Data Structures & Algorithms (88) | C Programming (94) | Java Advanced Programming (91)

Research & Project Experience

ZhaoXi (Drawsee) — AI Circuit Learning Platform

Apr. 2024 – Present

Project Leader | National-Level Innovation & Entrepreneurship Programme, BUPT | Key Project, BUPT Institute of Innovation and Entrepreneurship | drawsee.cn

Built a deployed vertical-domain AI system for electronics education, integrating RAG, LLM-based agent interaction, and multimodal data processing. The engineering problems encountered here directly motivated the two research extensions below.

- **RAG Pipeline:** Constructed an end-to-end RAG pipeline (BGE-M3 + Milvus) over course slides and lab documents. Retrieved domain context is injected into the base LLM as structured knowledge signals, improving vertical-domain QA accuracy by **~45%**.
- **Multimodal Processing:** Designed a cross-modal alignment layer unifying Netlist/SPICE formal descriptions, natural-language documents, and scanned lab reports into a shared embedding space prior to vector indexing.
- **Agent Interaction:** History-aware multi-turn dialogue agents with dynamic prompt templates bridging circuit formal language and natural language; LLM-based task routing across Q&A, simulation, and circuit analysis modes.
- **System Architecture:** Spring Boot MVC backend; RabbitMQ + Redis Stream for high-concurrency throughput; Dockerised deployment; end-to-end latency ≤ 1.5 s; human-evaluated accuracy 82%; 500+ users; 73% retention.
- **Awards:** 1st Prize (Beijing) — China Collegiate Computing Contest 2024; 2nd Prize (Beijing) — China International Innovation Competition 2025.

Building ZhaoXi revealed two concrete problems: flat-vector retrieval loses formal circuit structure in multimodal documents, and LLM-driven extraction into knowledge graphs introduces noise without any visibility into graph quality. The two projects below address each problem directly.

CircuitModalProcessor — Circuit-Domain Modal Processor for RAG-Anything

Feb. 2025 – Present

Independent Research | Built on HKUDS/RAG-Anything

RAG-Anything's native `ImageModalProcessor` captions images as prose, losing component values, net names, and topological connections — unavailable for entity-level retrieval in circuit-domain documents. This project extends the RAG-Anything `1+3+N` modal processor framework with a circuit-aware structured processing pipeline.

- **Method:** VLM captioning + netlist parsing + component-to-entity mapping produces `CircuitComponent` and `CircuitConnection` objects as first-class KG entities and edges, rather than opaque caption text.
- **Phase 1 Results (Op-Amp topic):** Circuit-focused QA (E2) — hybrid circuit-aware condition **8/8** successful answers vs. **3/8** under the mixed baseline. General domain QA (E1) — hybrid 8/8, naive 8/8. Gain reflects task completion reliability, not answer verbosity.
- **Phase 1 Results (BJT topic):** Circuit-focused QA (E2) — hybrid **8/8**; mix 7/8. General domain QA (E1) — hybrid 7/8, naive 8/8. Validates that the gain generalises beyond the op-amp domain.
- **Known Limitation:** A third topic (source transformation / superposition) hit a chunk-level LLM worker timeout (600 s), identifying robustness under concept-dense pages as the next bottleneck.
- **Phase 2:** Chunk-level extraction fallback; formal domain QA benchmark; circuit-terminology retrieval diagnostics.

NoiseFilter-RAG — Graph Noise Diagnosis and Confidence-Aware Retrieval

Jan. 2025 – Present

Independent Research | Built on HKUDS/LightRAG (EMNLP 2025)

LightRAG relies on LLM-driven entity extraction to build knowledge graphs. This extraction step hallucinates — introducing wrong edges and polluted retrieval context that the original paper's answer-level evaluation does not directly expose. This project diagnoses and governs that noise with a full-pipeline approach.

- **Method:** Three-signal confidence scoring engine (extraction frequency, cross-document consistency, semantic plausibility) integrated into LightRAG's insertion path; noise-aware retrieval filter (soft/hard edge suppression) injected into the retrieval path; synthetic noise injection benchmark for controlled evaluation.
- **Stage 1 Results:** Graph strict precision $0.15 \rightarrow \mathbf{0.22}$; lenient precision $0.50 \rightarrow \mathbf{0.57}$; pairwise answer win rate $0.4385 \rightarrow \mathbf{0.4615}$ (100-edge manual annotation benchmark, formal *mix* dataset). Source-grounded judge (ROC AUC $0.5413 \rightarrow 0.5497$) and relation directionality audit pipelines delivered.
- **Stage 2 (active):** Chunk-entity binding gate (P0) — intercept hallucinated edges before graph insertion; extraction prompt constraints (P0); relation directionality preservation (P1); NLI-based confidence signal recalibration (P1).

Engineering deliverables include reproducible dual-run pipelines, graph-quality sampling and manual labeling tooling, pairwise LLM judge evaluation, source-grounded judge, and relation directionality / source-binding audit scripts.

Multimodal Video Virality Prediction — Classification Study

Sept. 2024 – Dec. 2024

Course Research Project, BUPT

Designed and trained a binary classification model predicting YouTube video popularity from heterogeneous multimodal metadata (titles, tags, temporal signals, engagement counts).

- Built feature pipeline combining TF-IDF text features, StandardScaler-normalised numerical signals, and JSON categorical encodings.
- Trained multi-layer neural network (TensorFlow/Keras) with EarlyStopping and ReduceLROnPlateau callbacks; automatic class-imbalance compensation; strong generalisation on held-out test data; commended by supervising faculty.
- Evaluated with Accuracy, Precision, Recall, and F1; visualised training dynamics with matplotlib.

Awards & Honors

- ⋆ **1st Prize (Beijing) — China Collegiate Computing Contest** 2024
  Project Lead; system architecture design & AI module development

- ⋆ **2nd Prize (Beijing) — China International Innovation Competition** 2024
  Project Lead (ZhaoXi platform)

- ⋆ **2nd Prize — ICT Industry-Education Integration Innovation Contest** 2024
  Team Lead & Technical Director

- ⋆ **1st Prize — 10th National Undergraduate Physics Experiment Competition** 2024
  Core Member; designed Python data acquisition & processing pipeline

- ⋆ **Honorable Mention (H Award) — MCM/ICM Mathematical Modeling Competition** 2023
  Core Member; modeling design & implementation

Technical Skills

RAG & Knowledge Graphs	LightRAG, RAG-Anything, LangChain; BGE-M3, Milvus, Qdrant, NetworkX; confidence scoring; noise-aware retrieval; graph quality evaluation
Graph Learning	PyTorch Geometric; graph instruction tuning (HKUDS/GraphGPT); knowledge graph construction and auditing
LLM & Agents	Transformer architecture; LoRA/QLoRA fine-tuning (LLaMA-2, GLM-4); Qwen-7B inference optimisation; history-aware multi-turn dialogue agents; prompt engineering
Multimodal	Vision-language models (Seed1.5-VL); BGE-M3 dense embeddings; cross-modal alignment; VLM captioning pipelines
Evaluation	RAGAS; pairwise LLM judge; NLI grounding signal; manual annotation pipelines; ablation experiment design
ML Training	TensorFlow/Keras (DNNs, EarlyStopping, ReduceLROnPlateau, class imbalance); scikit-learn; Accuracy / Precision / Recall / F1 evaluation
Engineering	Spring Boot, Redis, RabbitMQ, Docker/Compose, Git, Linux (Ubuntu 24.04)
Languages	Python (proficient) | Java (proficient) | SQL (proficient) | Go (upper-int.) | C/C++ (intermediate) | TypeScript/React (intermediate)

Research Statement

I study failure modes in RAG systems that production-level evaluation tends to paper over — graph noise from LLM entity extraction, structural information loss in multimodal document ingestion, and the gap between answer-quality metrics and what actually goes wrong inside the retrieval graph. My two ongoing independent research projects (NoiseFilter-RAG and CircuitModalProcessor) both extend open-source work from the HKUDS Data Intelligence Lab and are grounded in concrete problems I first encountered while building ZhaoXi, a deployed AI platform for electronics education at BUPT.

I am broadly interested in how structured knowledge representations — knowledge graphs, graph foundation models, and graph-enhanced retrieval — can make LLM systems more reliable, interpretable, and domain-aware. I am also drawn to LLM-based agent systems, particularly how tool-augmented agents can ground their reasoning in retrieved structured knowledge rather than relying solely on parametric memory. I am seeking a research position where I can contribute to work in these directions while deepening my theoretical foundations under faculty mentorship.